

# Multiclass Classification and Polynomial Regression

A Comparative Machine Learning Study on MNIST and California Housing

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## Executive Summary

This report documents the design, implementation and critical evaluation of four supervised machine-learning models applied to two benchmark problems. Three classifiers — k-Nearest Neighbours (kNN), a Support Vector Machine (SVM) across linear, RBF and polynomial kernels, and a feed-forward Neural Network — were trained on the MNIST handwritten-digit dataset and compared on accuracy, precision, recall and F1-score. A Polynomial Regression model was then applied to the California Housing dataset across multiple degrees and assessed using MSE, RMSE and  $R^2$ . The implementation follows established machine-learning practice (Géron, 2022) using the scikit-learn library (Pedregosa et al., 2011).

The neural network achieved the strongest classification performance while retaining the fast inference required for deployment; the RBF-kernel SVM delivered the best balance of accuracy against computational cost on a constrained compute budget; and kNN, although simple and interpretable, scaled poorly to the high-dimensional pixel space. For the regression task, the simplest model (degree 1) provided by far the best generalisation, while higher polynomial degrees overfit catastrophically — a clear demonstration that complexity without regularisation harms performance. The study reinforces a central data-engineering principle: model selection is a context-dependent trade-off between accuracy, latency, scalability, interpretability and compute cost, rather than a search for a single universally optimal algorithm.

## 1. Introduction

Classification and regression are the two foundational tasks of supervised machine learning and underpin a large proportion of production data systems. This project applies both to industry-representative benchmarks. MNIST, a balanced ten-class image dataset of handwritten digits, is the standard testbed for multiclass classification (LeCun et al., 1998); California Housing, originally compiled by Pace and Barry (1997), is a continuous-target regression problem analogous to property valuation and demand forecasting.

The objectives are twofold. First, to implement and critically compare three classification algorithms on MNIST, explaining the role of each parameter and metric rather than reporting results in isolation. Second, to implement polynomial regression on California Housing and analyse the bias–variance trade-off as model complexity increases. Throughout, results are interpreted and connected to real-world data-engineering practice.

## 2. Methodology

All experiments were implemented in Python using scikit-learn (Pedregosa et al., 2011) for kNN, SVM and regression, and TensorFlow/Keras for the neural network. To ensure reproducibility and satisfy the unit's academic-integrity requirements, every stochastic operation used a fixed random seed equal to the last two digits of the student ID, and dataset variables were named using the student's first and last name.

MNIST was loaded via `fetch_openml` (70,000 images, 784 features each) and split 80/20 with stratification to preserve class balance. Because kernel SVMs scale between quadratic and cubic in the number of samples, the SVM comparison was conducted on a stratified 10,000-image

subset — a standard engineering compromise that keeps the kernel comparison fair while remaining computationally tractable on a CPU. kNN and the neural network were evaluated on the full dataset. For regression, the California Housing data was standardised and expanded with polynomial features inside a scikit-learn pipeline to prevent data leakage, a common but easily overlooked source of over-optimistic results (Géron, 2022).

### 3. Dataset Description

#### 3.1 MNIST

MNIST contains 70,000 greyscale images of handwritten digits, each 28×28 pixels flattened into a 784-dimensional vector with intensities from 0 to 255. The ten classes (digits 0–9) are approximately uniformly distributed, which justifies accuracy as a headline metric while still motivating per-class precision, recall and F1 to detect systematic weaknesses on visually similar digit pairs (e.g. 4 and 9).

#### 3.2 California Housing

The California Housing dataset comprises 20,640 census-block records with eight numeric features — median income, house age, average rooms and bedrooms, block population, average occupancy, latitude and longitude. The target is the median house value expressed in units of \$100,000 and is capped at \$500,000, an important artefact that shapes the regression results.

### 4. Model Implementation

#### 4.1 k-Nearest Neighbours

kNN is a non-parametric, instance-based learner that classifies a query by majority vote among its k nearest training points under Euclidean distance. The model was configured with k=10, distance weighting so that nearer neighbours contribute more strongly, and parallelised neighbour search. Its principal limitation is computational: prediction requires comparing each query against the entire training set, and distance becomes less discriminative in 784 dimensions — the curse of dimensionality (Géron, 2022).

#### 4.2 Support Vector Machine

The SVM, introduced by Cortes and Vapnik (1995), seeks the maximum-margin separating hyperplane and uses the kernel trick to handle non-linearly separable classes. Three kernels were evaluated. The C parameter controls margin softness — the bias–variance trade-off between tolerating misclassification and fitting the training data tightly — while gamma governs the reach of each training point's influence for the RBF and polynomial kernels. Training time was recorded alongside accuracy to quantify the cost of each kernel.

#### 4.3 Neural Network

A feed-forward network with two hidden layers (256 and 128 units, ReLU activation) and dropout regularisation was implemented, terminating in a ten-unit softmax output. The Adam optimiser (Kingma & Ba, 2015) and categorical cross-entropy loss were selected as the standard, robust choices for multiclass classification, and dropout was applied to reduce overfitting by preventing

neuron co-adaptation (Srivastava et al., 2014). Pixel values were normalised to the unit interval to keep gradients well-scaled and accelerate convergence.

#### 4.4 Polynomial Regression

Polynomial regression extends linear regression by generating powers and interactions of the input features, enabling a linear model to fit curved relationships. Degrees one through four were fitted within a pipeline that first standardised the features — essential because high-degree powers of unscaled features produce numerically unstable magnitudes.

### 5. Results and Analysis

The classification results from my executed notebook are summarised below.

| Model          | Accuracy | Precision | Recall | F1-score |
|----------------|----------|-----------|--------|----------|
| kNN (k=10)     | 0.9716   | 0.9722    | 0.9712 | 0.9716   |
| SVM (linear)   | 0.9145   | 0.9151    | 0.9136 | 0.9137   |
| SVM (RBF)      | 0.9620   | 0.9620    | 0.9619 | 0.9619   |
| SVM (poly)     | 0.9595   | 0.9597    | 0.9591 | 0.9593   |
| Neural Network | 0.9784   | 0.9782    | 0.9782 | 0.9782   |

The neural network achieved the highest classification performance (accuracy 0.9784, F1 0.9782), narrowly ahead of kNN (0.9716). Across SVM kernels, the RBF kernel clearly outperformed the linear kernel (0.9620 versus 0.9145) because MNIST digit classes are not linearly separable in pixel space, while the linear kernel trained fastest (10.3 s versus 15.5 s for RBF). The neural network combined top accuracy with the fastest inference, making it the most deployment-ready model. The regression results below show how performance varied with polynomial degree.

| Degree | MSE    | RMSE  | R <sup>2</sup> |
|--------|--------|-------|----------------|
| 1      | 0.508  | 0.712 | 0.609          |
| 2      | 6.39   | 2.53  | -3.92          |
| 3      | 21833  | 147.8 | -16800         |
| 4      | 7.86e8 | 28044 | -6.05e8        |

### 6. Critical Discussion

The comparison exposes clear trade-offs. kNN requires no training but pays a heavy cost at every prediction and scales poorly in both memory and time; it is best suited to small problems where interpretability (nearest-example explanations) is valuable. The SVM is costly to train but fast to query and remains highly competitive on medium-sized, high-dimensional data such as text. The neural network trains once and serves predictions extremely cheaply, scaling gracefully to industrial volumes — the decisive property for production image pipelines.

For regression, the polynomial-degree experiment produced a striking demonstration of overfitting. Degree 1 gave the best result ( $R^2 = 0.61$ ,  $RMSE = 0.71$ ), while degrees 2, 3 and 4 collapsed catastrophically to large negative  $R^2$  values ( $-3.9$ ,  $-16,800$  and roughly  $-605$  million respectively). Expanding eight features into high-degree powers and interaction terms created so many parameters that the unregularised linear regression fit noise and became numerically unstable, predicting far worse than the mean. The key insight is that greater model complexity does not guarantee better performance: without regularisation, high-degree polynomial regression suffers a variance explosion, and the appropriate choice here is the simplest model (degree 1) or a regularised alternative such as Ridge regression. The most natural improvement for MNIST is a convolutional neural network, which exploits spatial structure and routinely exceeds 99% accuracy (Géron, 2022); for housing, gradient-boosted trees or regularised regression typically outperform plain polynomial regression.

If developing a production machine-learning system for digit recognition at scale, the recommended model would be the neural network (ideally a convolutional neural network), because it combines the highest classification accuracy with very fast inference and the ability to scale gracefully to large data volumes — the decisive properties for a deployed system, even though it sacrifices the interpretability that makes kNN attractive for smaller problems.

## 7. Data Ethics Discussion

Even benchmark datasets raise ethical considerations that become material upon deployment. MNIST was collected from a narrow US population, so its handwriting styles are not globally representative and a model trained on it may systematically misread other regional conventions — bias inherited directly from the data.

- **Fairness:** incorrect predictions carry uneven, real-world costs (misdirected payments, lost mail, incorrect grades), so error rates should be audited per subgroup rather than only in aggregate.
- **Privacy:** handwriting is weakly biometric, so large-scale recognition systems require consent, data minimisation and secure storage consistent with privacy regulation.
- **Responsible deployment:** confidence calibration, human-in-the-loop review of low-confidence cases, diverse training data, and transparent documentation (model cards) mitigate these risks.

## 8. Conclusion

This study implemented and critically compared three classifiers on MNIST and a polynomial regressor on California Housing. The neural network offered the best combination of accuracy and deployability, the RBF SVM the best accuracy-to-cost balance under compute constraints, and kNN the greatest simplicity at the expense of scalability. For regression, the degree-1 model performed best while higher degrees overfit dramatically, demonstrating that increasing complexity without regularisation degrades performance. The overarching conclusion is that effective machine-learning engineering is the disciplined matching of model to context — data scale, latency, interpretability and compute budget — supported by rigorous evaluation and conscientious attention to data ethics.

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